Identification of
Seed-Borne Fungi

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Seed Pathology

Definition & Objectives

Definition

Seed pathology is the scientific study of diseases and pathogens associated with or transmitted through seeds.
It encompasses their detection, mechanisms of infection, epidemiology, and control strategies — critical for ensuring food security and preventing the global spread of plant diseases.

Seed Health Testing Laboratory

Objectives of Seed Pathology

Disease Detection & Identification

Accurate detection of seed-borne pathogens to prevent disease outbreaks.

Preventing Disease Spread

Quarantine and regulatory measures to halt pathogen dissemination.

Epidemiological Research

Studying disease development and host-pathogen interactions.

Management & Control

Developing seed treatment and cultural control strategies.

Seed Quality Assurance

Ensuring seeds meet health standards for optimum germination & yield.

Trade & Regulatory Support

Facilitating international seed trade by certifying seed health.

Methods of Seed Health Testing

Overview of Standard Procedures

1

Dry Seed Inspection

Visual examination of seeds for discolouration, deformation, or fungal growth without any pre-treatment.

2

Washing Test / Embryo Count

Microscopic examination of the suspension obtained by washing seeds; includes whole embryo count for systemic pathogens.

3

Incubation Tests

Blotter, Agar Plate, Deep-Freezing Blotter, Water Agar, and Test Tube Agar methods to promote pathogen sporulation.

4

Seedling Symptom Test

Hiltner's brick-stone, Sand, Standard Soil, and Test Tube Agar methods to assess disease expression in germinating seeds.

5

Growing-On Test

Seeds are sown and plants grown to maturity under controlled conditions to observe systemic disease symptoms.

6

Serological Tests

ELISA and related immunological techniques for rapid, specific detection of seed-borne viruses and bacteria.

7

Indicator Plant Test

Inoculation of a susceptible indicator host to confirm the presence and viability of a specific pathogen.

8

Electron Microscopy

Ultra-structural examination to identify viruses and other sub-microscopic pathogens associated with seeds.

Blotter Method

Standard Incubation Procedure for Seed Health Testing

01

Preparation

Two layers of blotter paper are placed in sterile Petri dishes and thoroughly soaked with distilled water to maintain high moisture throughout the incubation period.

02

Surface Sterilization (Optional)

Seeds may be surface-sterilized with 1% NaOCl for 1 minute and rinsed with sterile distilled water to suppress saprophytic contaminants.

03

Seeding

20 seeds per Petri dish are placed equidistant on the moist blotter paper under aseptic conditions. Two replications are maintained for each seed lot.

04

Incubation

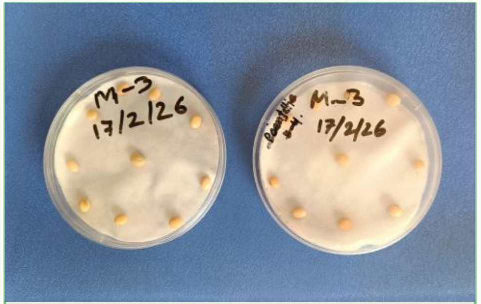
Petri dishes are incubated at room temperature for 7 days. Adequate moisture is maintained throughout.

05

Examination

After 7 days, seeds and associated fungal growth are examined under a stereoscopic microscope and compound microscope for pathogen identification.

Seeding on blotter paper



Labelled Petri dishes (M-3, 17.02.2026)

Results of Blotter Method

Soybean Seed Health Assessment – 20 Seeds Tested

Fig: Petri dishes after 7-day incubation
(Seeds numbered 1–20)

Observation Summary

Seeds No. 4, 11, 14, 20
Healthy & Viable ✓

Seeds No. 1, 6, 7, 19
Infected with Alternaria sp. ⚠

Seeds No. 2, 12, 17
Infected with Aspergillus sp. ⚠

Seeds No. 5, 13, 16
Infected with Fusarium sp. ⚠

Seeds No. 15
Infected with Rhizopus sp. ⚠

Seeds No. 3, 8, 9, 10, 18
Non-viable (No germination) ✗

Pathogen Description — *Alternaria sp.*

Seeds Infected: No. 1, 6, 7, 19

Microscopic view of Alternaria sp.

Systematic Position

Kingdom	:	Fungi
Division	:	Eumycota
Sub-division	:	Deuteromycotina
Class	:	Hyphomycetes
Order	:	Moniliales
Family	:	Dematiaceae
Genus	:	*Alternaria*
Species	:	*Alternaria sp.*

Morphology

Dark, club-shaped conidia with transverse AND longitudinal septa — distinctive muriform (brick-like) appearance. Conidiophores are simple or branched, dark-coloured.

Diseases Caused

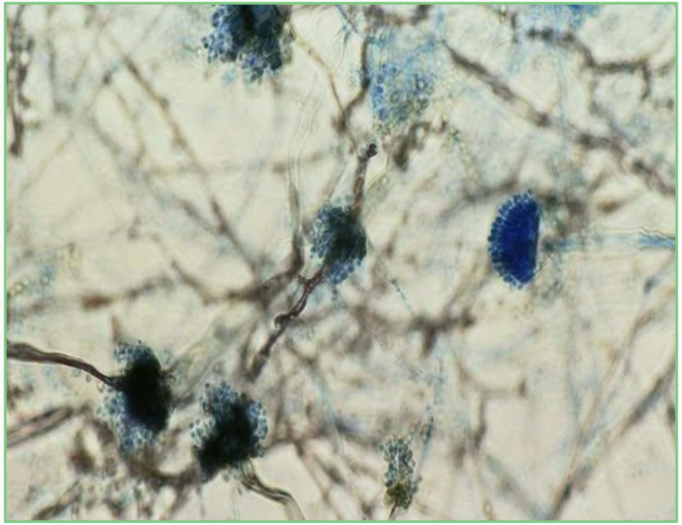
Alternaria Leaf Spot (Alternaria sojina), Seedling Blight, Collar Rot, Pod & Seed Rot. Causes significant reductions in seed quality and germination percentage.

Control Measures

Seed treatment: Thiram @ 2.5 g/kg seed. Propiconazole and Carbendazim are also effective. Crop rotation and destruction of infected crop debris recommended.

Pathogen Description — *Aspergillus sp.*

Seeds Infected: No. 2, 12, 17



Microscopic view of Aspergillus sp.

Systematic Position

Kingdom	:	Fungi
Division	:	Eumycota
Sub-division	:	Ascomycotina
Class	:	Plectomycetes
Order	:	Aspergillales
Family	:	Aspergillaceae
Genus	:	*Aspergillus*
Species	:	*Aspergillus sp.*

Morphology

Globose to sub-globose vesicle at the apex of a long, non-septate conidiophore bearing radiating chains of pigmented conidia (blue-green to black). Foot cell present at the base.

Diseases Caused

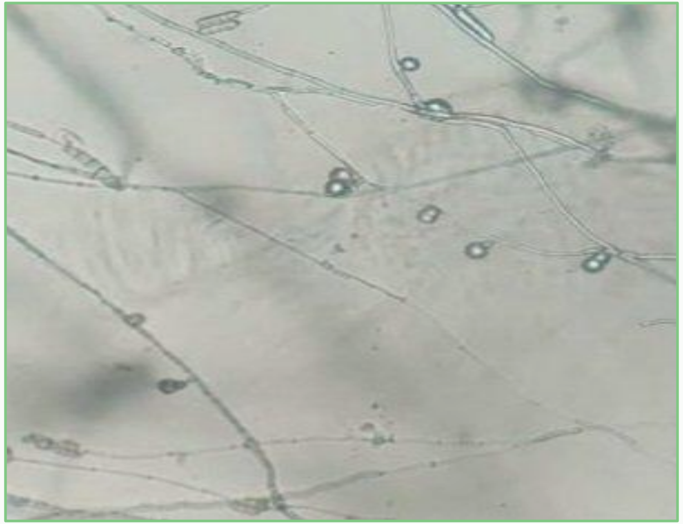
Aspergillus Seed Rot (black/yellow mould), Aspergillus Crown Rot, storage rots, aflatoxin contamination, and germination failure of stored seeds.

Control Measures

Store seeds at low moisture (<13%) and cool temperature. Thiram @ 2.5 g/kg seed. Proper seed drying before storage is critical to prevent Aspergillus proliferation.

Pathogen Description — *Fusarium sp.*

Seeds Infected: No. 5, 13, 16



Microscopic view of Fusarium sp.

Systematic Position

Kingdom	:	Fungi
Division	:	Eumycota
Sub-division	:	Deuteromycotina
Class	:	Hyphomycetes
Order	:	Moniliales
Family	:	Tuberculariaceae
Genus	:	*Fusarium*
Species	:	*Fusarium sp.*

Morphology

Macroconidia: canoe/banana-shaped, multi-septate (3–7). Microconidia: oval, 0–1 septate, borne on simple or branched conidiophores. Chlamydospores may form in older cultures.

Diseases Caused

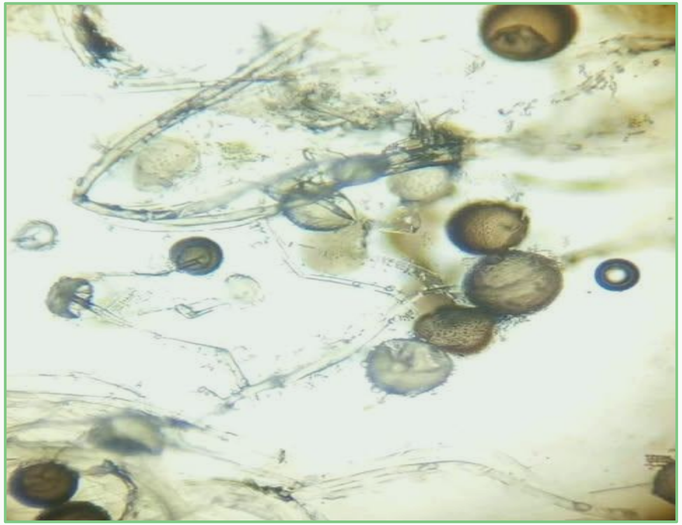
Fusarium Root Rot, Damping-off, Fusarium Wilt (Sudden Death Syndrome), Seed Decay, and Seedling Blight. Causes vascular browning and sudden plant death in soybeans.

Control Measures

Thiram @ 2.5 g/kg seed. Metalaxyl + Fludioxonil combinations are effective. Avoid soil compaction. Crop rotation with non-host plants reduces inoculum levels.

Pathogen Description — *Rhizopus sp.*

Seed Infected: No. 15



Microscopic view of Rhizopus sp.

Systematic Position

Kingdom	:	Fungi
Division	:	Eumycota
Sub-division	:	Zygomycotina
Class	:	Zygomycetes
Order	:	Mucorales
Family	:	Mucoraceae
Genus	:	*Rhizopus*
Species	:	*Rhizopus sp.*

Morphology

Coenocytic (non-septate) mycelium, fast-growing and cottony white, turning dark grey to black with maturity. Long sporangiophores arise from rhizoids and terminate in large globose sporangia packed with spores.

Control Measures

Thiram @ 2.5 g/kg seed. Maintain low storage humidity (<13% MC) and temperature. Good field drainage and proper post-harvest drying are the most effective preventive measures.

Diseases Caused

Seed Rot & Soft Rot in soybean, characterised by rapid watery tissue collapse covered in white fluffy growth turning black. Also causes post-harvest decay and dampness-related storage losses.

Comparative Summary of Identified Pathogens

Soybean Seed-Borne Fungi — Blotter Method Results

Pathogen	Seeds No.	Sub-division	Key Morphology	Primary Disease	Control
Alternaria sp.	1, 6, 7, 19	Deuteromycotina	Muriform conidia, dark, club-shaped	Leaf Spot, Seedling Blight	Thiram 2.5 g/kg
Aspergillus sp.	2, 12, 17	Ascomycotina	Vesicle, radiating conidia chains	Black Mould, Crown Rot	Thiram 2.5 g/kg + dry storage
Fusarium sp.	5, 13, 16	Deuteromycotina	Canoe-shaped macroconidia	Root Rot, Damping-off, Wilt	Thiram + Metalaxyl
Rhizopus sp.	15	Zygomycotina	Coenocytic, sporangiophore with rhizoids	Seed Rot, Soft Rot	Thiram + dry storage
Healthy & Viable	4, 11, 14, 20	—	No fungal growth observed	—	—
Non-viable	3, 8, 9, 10, 18	—	No germination, no pathogen	—	—

Note: All fungicide treatments are seed dressings. Integrated Disease Management (IDM) is recommended alongside chemical control.

Seed Health Testing — Advantages & Disadvantages

With Special Reference to the Blotter Method

✓ Advantages

- Rapid pathogen detection by visual observation of growth characters after incubation.
- Economical and simple — requires no specialised or expensive equipment.
- Combined in vitro and in vivo approach yields highly reliable results.
- Pathogenic potential can be gauged from signs and symptoms on germinating seedlings.
- Applicable to a wide range of fungal pathogens across all seed types.
- Widely adopted standard method (ISTA); preferred over agar plate where needed.

✗ Disadvantages

- Fast-growing fungi (e.g., Rhizopus, Mucor) may overgrow and mask slow-growing pathogens.
- Pathogenic bacteria and most viruses cannot be detected by this method.
- Time-consuming: requires 7–10 days incubation before results are available.
- Pathogenicity of detected organisms cannot be confirmed directly.
- Very small seeds or those with high surface contamination may yield ambiguous results.

Conclusion

1. Four seed-borne fungal pathogens — *Alternaria, Aspergillus, Fusarium*, and *Rhizopus* — were successfully identified from 20 soybean seeds using the standard blotter method.

2. Only 4 of 20 seeds (20%) were healthy and viable; 5 seeds (25%) were non-viable, highlighting a serious seed health concern.

3. The blotter method proved to be a reliable, economical and effective tool for seed health testing in a teaching laboratory setting.

4. Seed treatment with Thiram @ 2.5 g/kg is recommended for all identified pathogens. Proper storage conditions are equally critical.

5. Regular seed health testing before sowing is essential to prevent introduction and spread of seed-borne pathogens into production fields.